Department of Computer Science And Engineering (CSE)
BRAC University

CSE 251 – Electronic Devices and Circuits

LECTURE: ##

MOSFET

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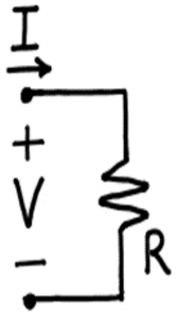
OUTLINE

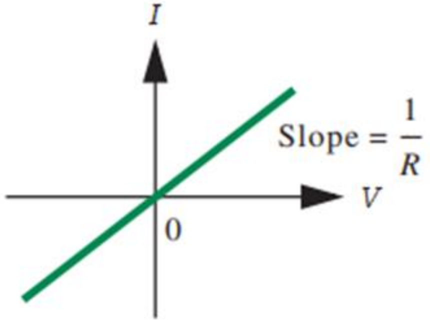
- Three Terminal Devices
- Introduction to Electronic Switches
- Introduction to Controlled Sources
- Introduction to MOSFET
- MOSFET - S Model
- MOSFET as Digital Switch
- Designing Logic Gates with MOSFETs

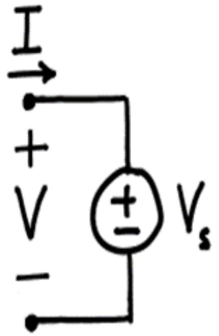
Two Terminal Devices (Review)

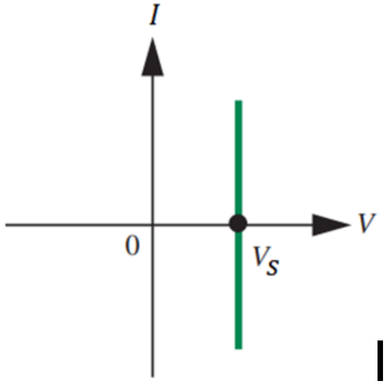
IV Characteristics of two terminal devices

Linear Devices

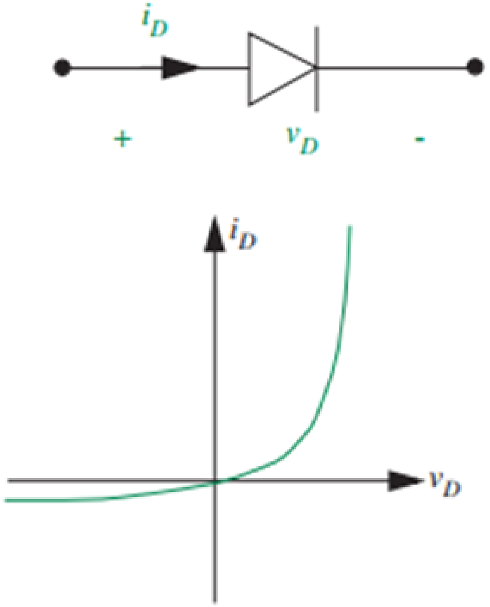








Non-Linear Devices



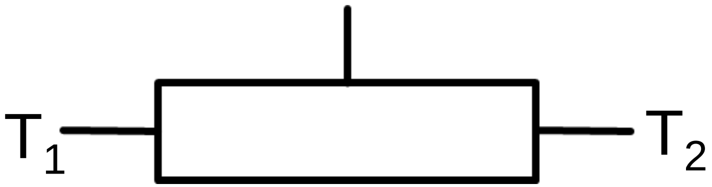
IV Characteristics of two terminal devices are **fixed** until the device structure and material properties are changed

Three Terminal Devices

IV of two terminal can be controlled using a third terminal.
Example: **Switch, MOSFET, BJT**

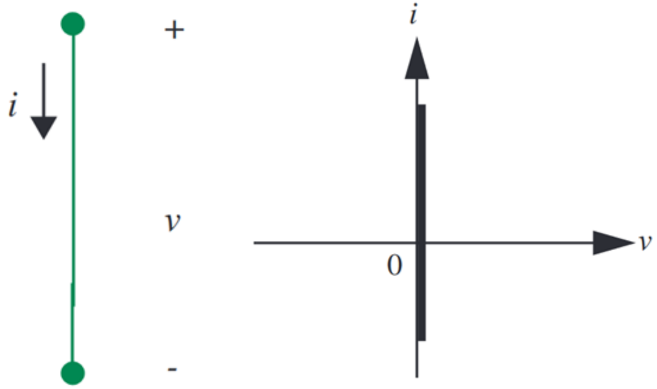
Switch

C → Control signal (physical/Voltage/Current)

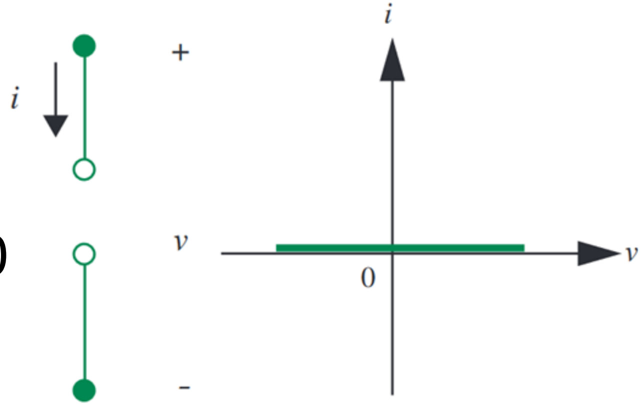


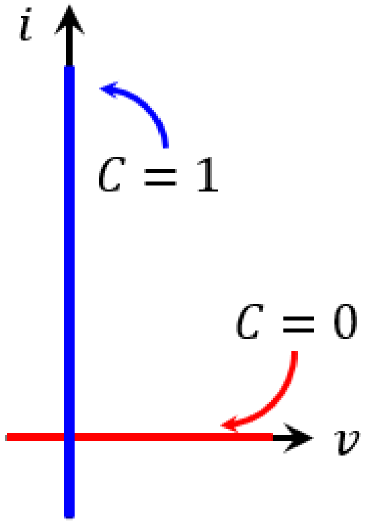
IV characteristics of T_1 and T_2 can be controlled by **C**

C=1



C=0





Switch ON (Short circuit)

Switch OFF(Open circuit)

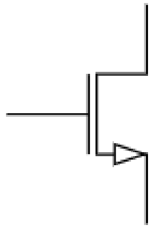
Switch - Types

Switch can be classified based on their method of control-

- ❑ **Analog:** Controlled manually/physically (button, slider, knob)
- ❑ **Digital :** Controlled by electrical signal (current, voltage)

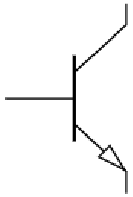
Digital Switch

Voltage controlled



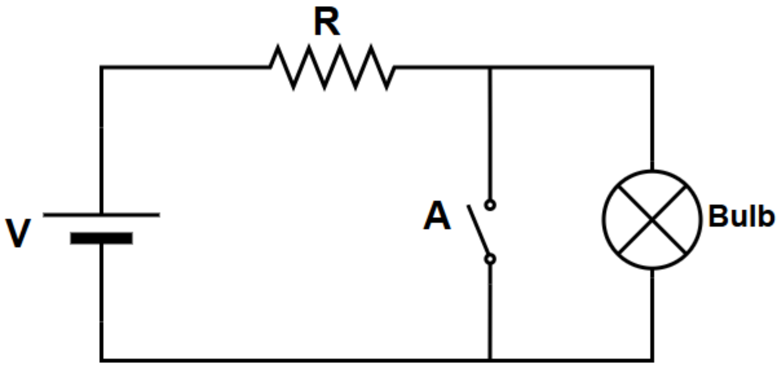
MOSFET

Current controlled



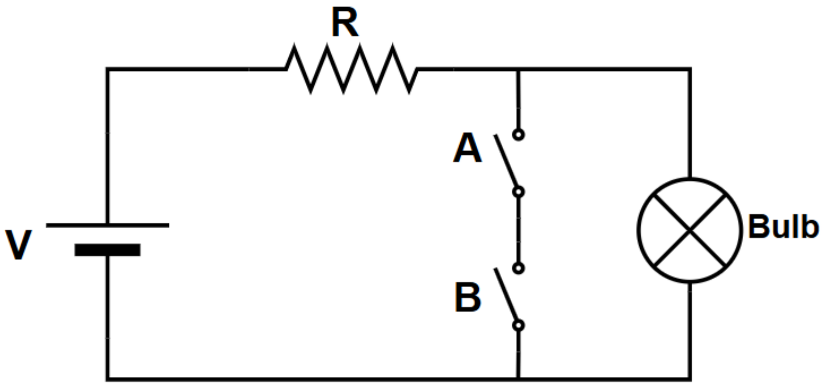
BJT

Switch Application - Logic Gates



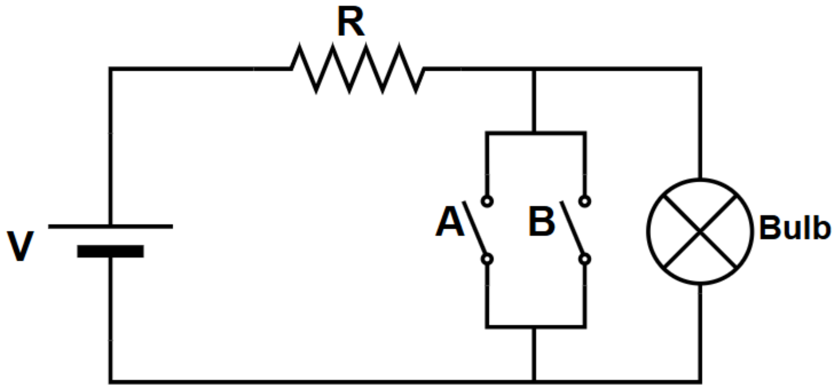
A	Bulb
0	ON
1	OFF

NOT Operation (inverter)



A	B	Bulb
0	0	ON
0	1	ON
1	0	ON
1	1	OFF

NAND Operation



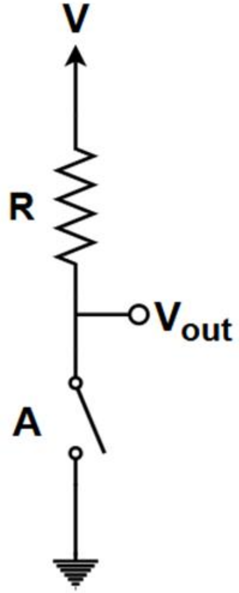
A	B	Bulb
0	0	ON
0	1	OFF
1	0	OFF
1	1	OFF

NOR Operation

Switch Application - Logic Gates

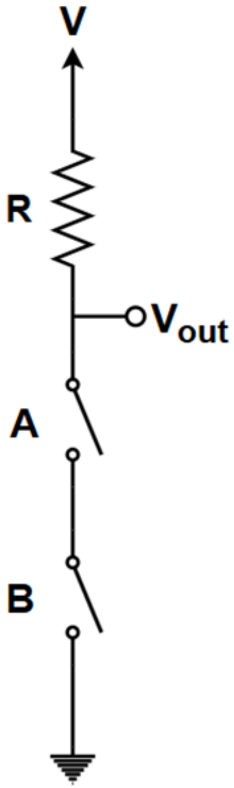
- Alternative Representation

NOT Gate



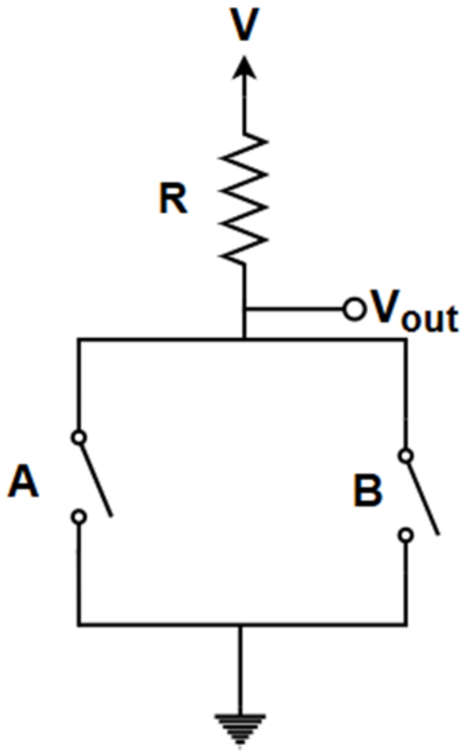
$$V_{out} = \overline{A}$$

NAND Gate



$$V_{out} = \overline{AB}$$

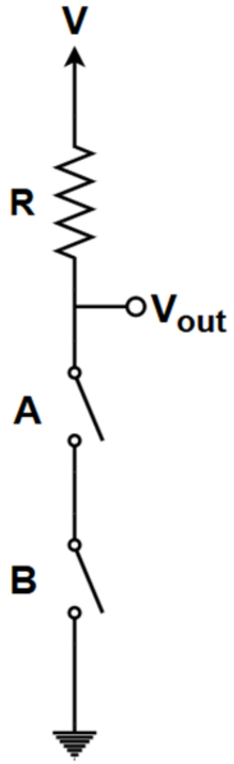
NOR Gate



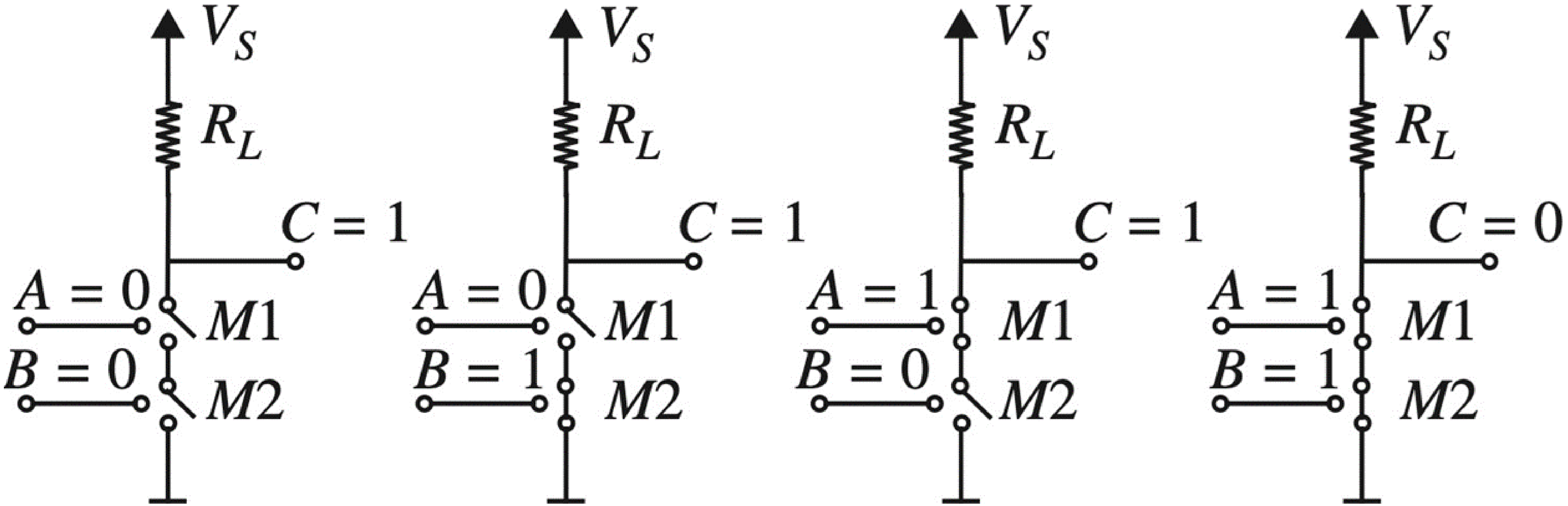
$$V_{out} = \overline{A + B}$$

Switch Application - Logic Gates

- 2- Input NAND Gate

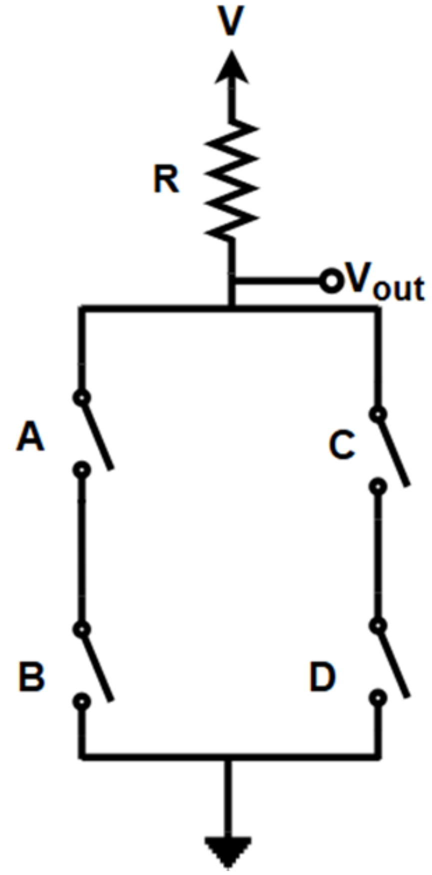


$$V_{out} = \overline{AB}$$



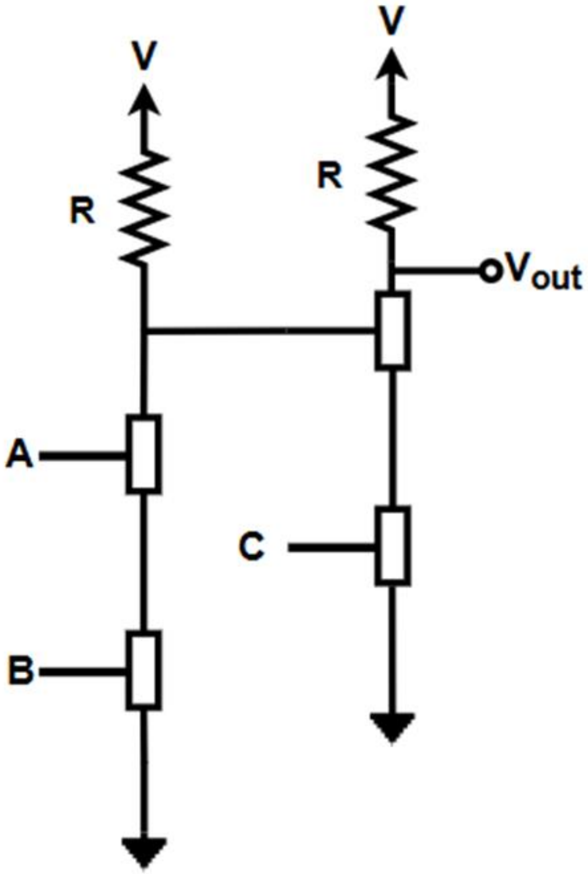
Examples

1.



$$V_{out} = \overline{AB + CD}$$

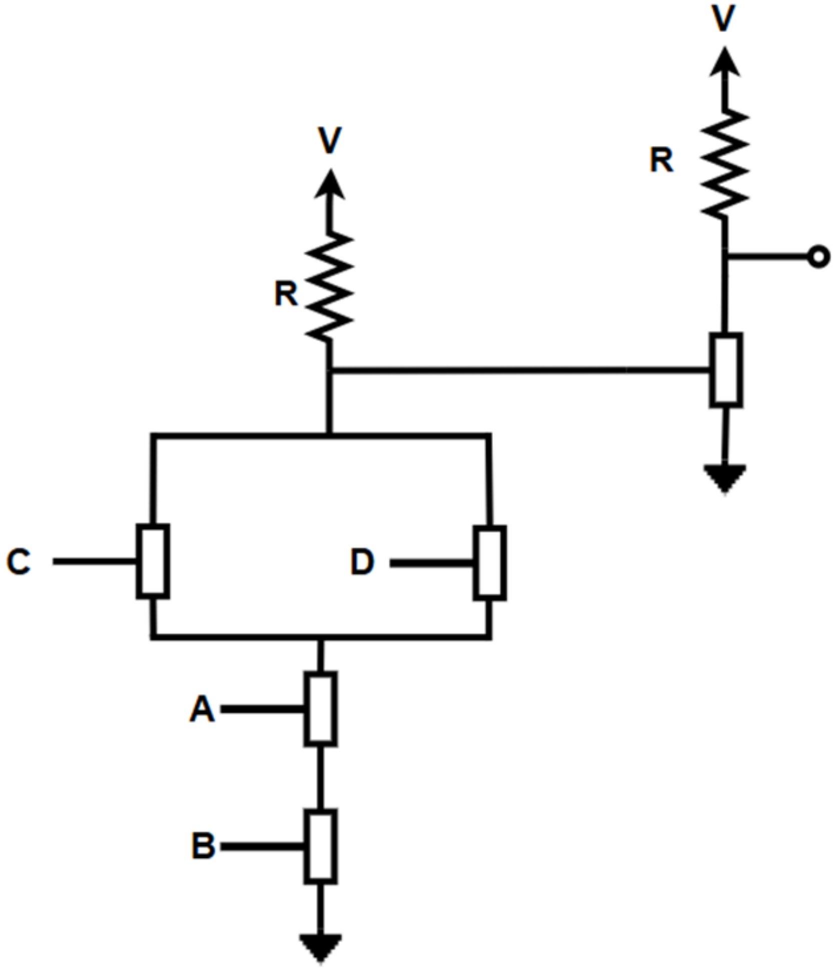
2.



$$V_{out} = \overline{\overline{AB}.C}$$

Examples

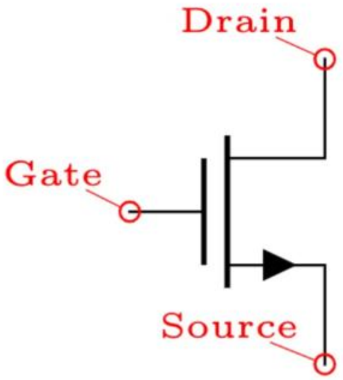
- Implement using switches: f = A.B.(C+D)



$$V_{out} = \overline{\overline{A.B.(C+D)}}$$
$$= A.B.(C+D)$$

Transistors in Digital Switching Application

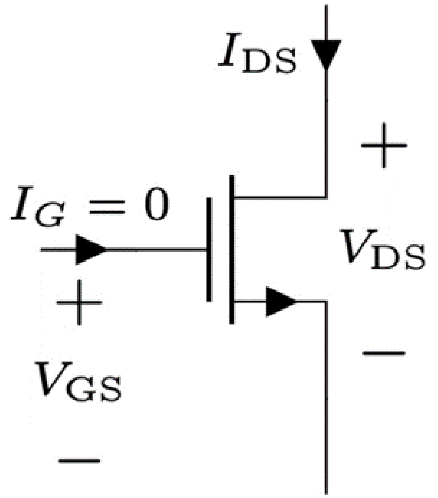
- ❑ A Transistor is a three terminal semiconductor device that is more useful than two terminal devices as it can be used in a wide range of applications - from signal amplification to digital logic and memory.
- ❑ Two main types-
  - ❖ Voltage Controlled → MOSFET (gate voltage controls switching)
  - ❖ Current Controlled → BJT (base current controls switching)
- ❑ MOSFET is a **voltage controlled** transistor widely used in digital circuits.
- ❑ **MOSFET**→ **M**etal **O**xide **S**emiconductor **F**ield **E**ffect **T**ransistor
- ❑ Three Terminals :
  - ❖ Gate (G)
  - ❖ Drain (D)
  - ❖ Source (S)



Circuit Symbol

MOSFET - S Model

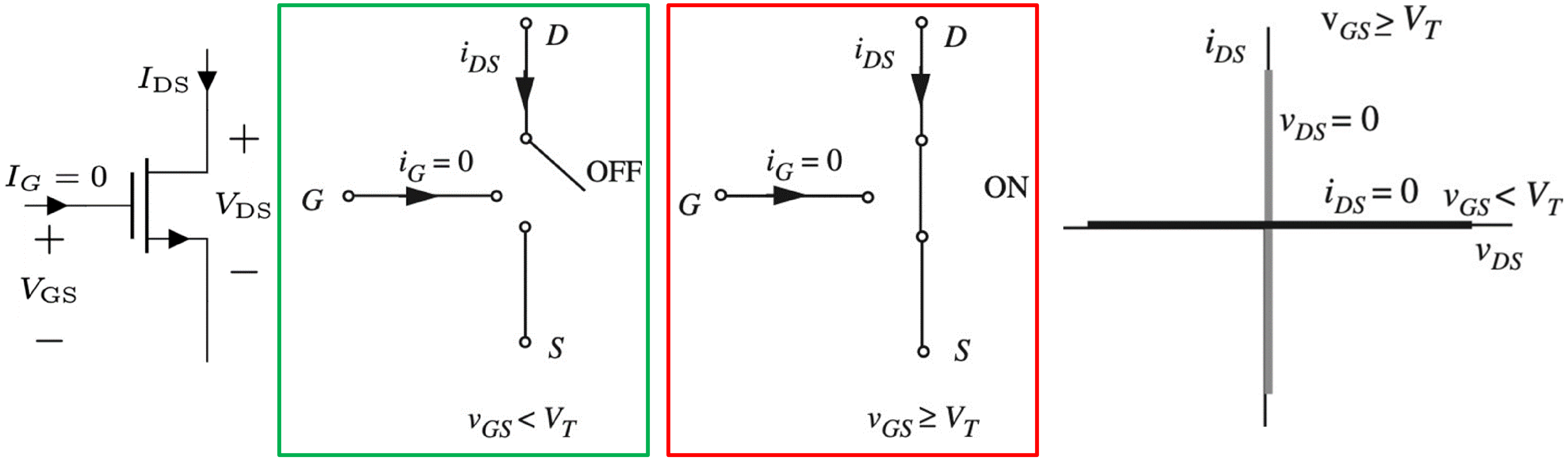
- ❑ The **Switch model** represents a MOSFET as an **ideal ON/OFF switch** used in digital circuits.
- ❑ Depending on the **gate to source voltage (V_{GS})** , the MOSFET either conduct current (ON) or block current (OFF).
- ❑ **Threshold Voltage V_{TH}** : The minimum **gate to source voltage (V_{GS})** to turn the MOSFET ON. It depends on device structure (doping and oxide thickness), applied voltages and temperature.
- ❑ For N-channel MOSFET:
  - ❖ $V_{GS} < V_{TH} \rightarrow$ OFF state (open circuit)
  - ❖ $V_{GS} \geq V_{TH} \rightarrow$ ON State (short circuit)



➢ Note: This is an approximation - real MOSFETs have resistance, threshold variations and leakage current.

MOSFET S Model- IV Characteristics

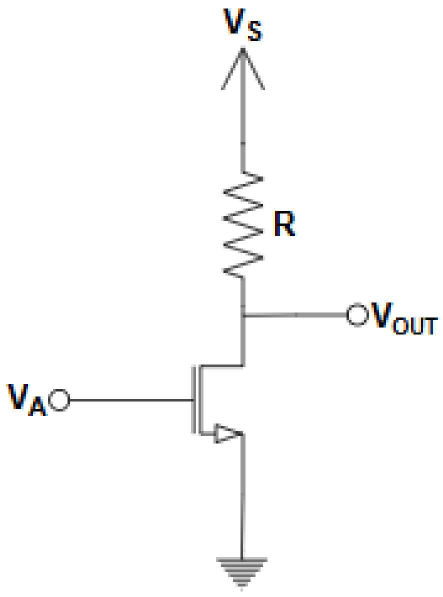
❑ The IV characteristics (I_{DS} vs V_{DS}) depends on V_{GS}



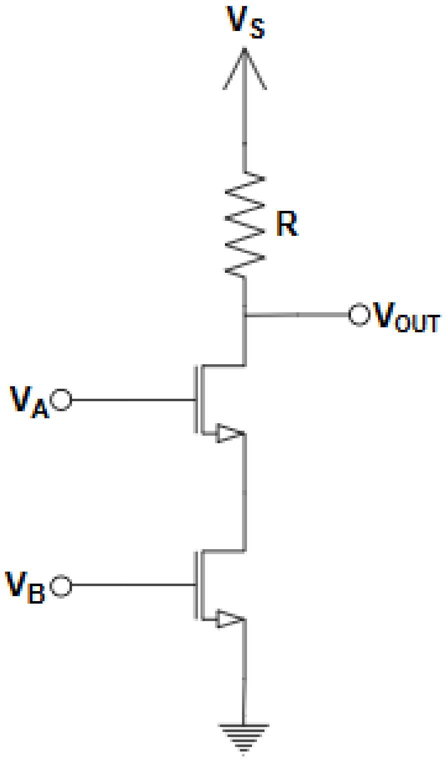
➢ Note: This is an approximation - real MOSFETs have resistance, threshold variations and leakage current.

Logic Gate Using MOSFETs

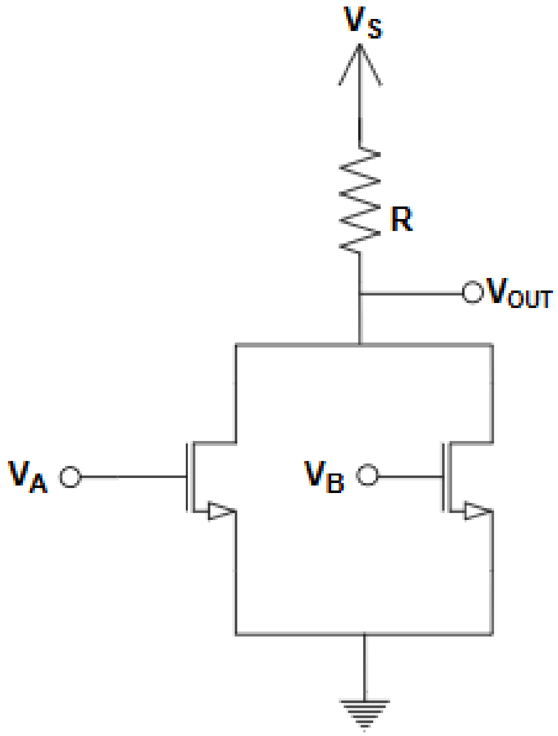
- According to the S-Model , MOSFET (approximately) behave like a switch. So logic functions can be implemented using MOSFETs.
- Just replace the switches with MOSFETs.



NOT Gate (inverter)



NAND Gate

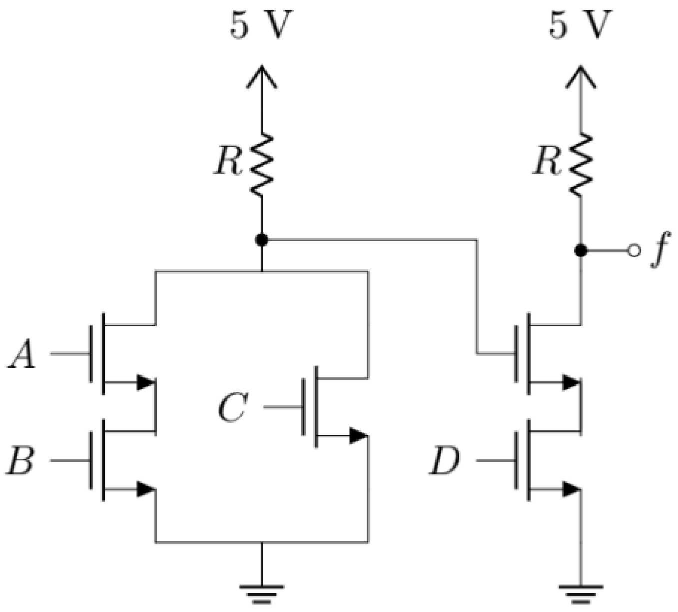


NOR Gate

MOSFET Logic Gate - Examples

Example 1

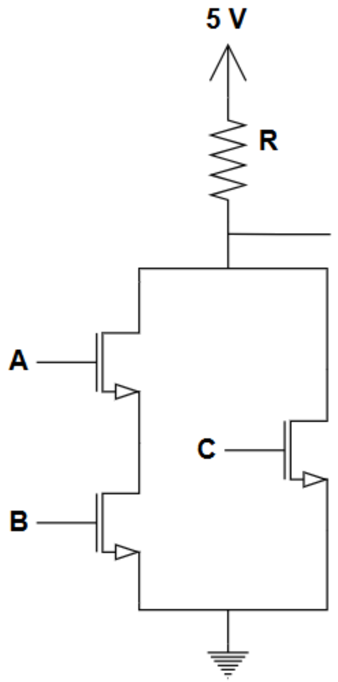
- Analyze the circuit-1 to find *f* in terms of Boolean inputs A, B, C and D



Circuit -1

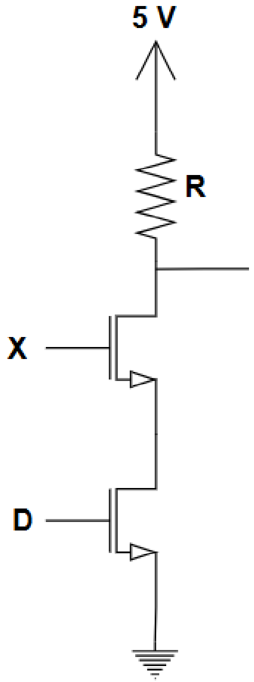
Solution:

Output of the first stage is X



$$X = \overline{AB + C}$$

Output of the final stage is *f*



$$f = \overline{X.D}$$

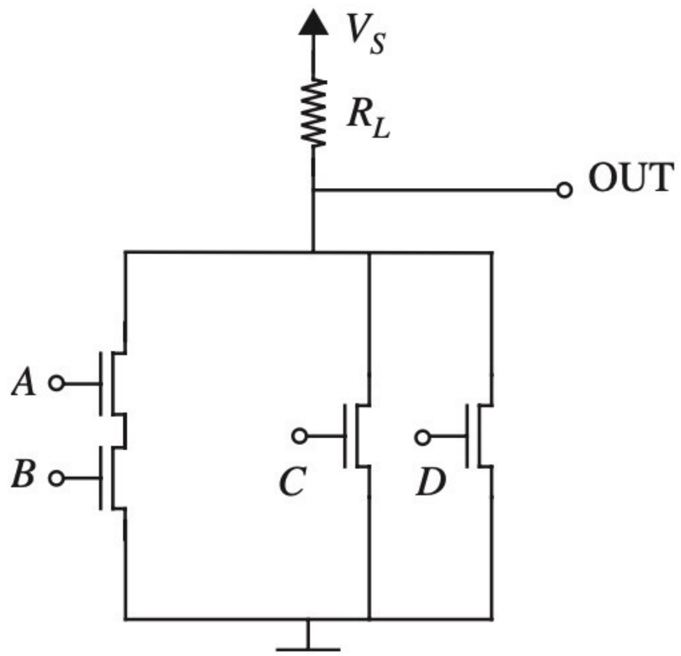
$$f = \overline{(\overline{AB + C}).D}$$

MOSFET Logic Gate - Problems

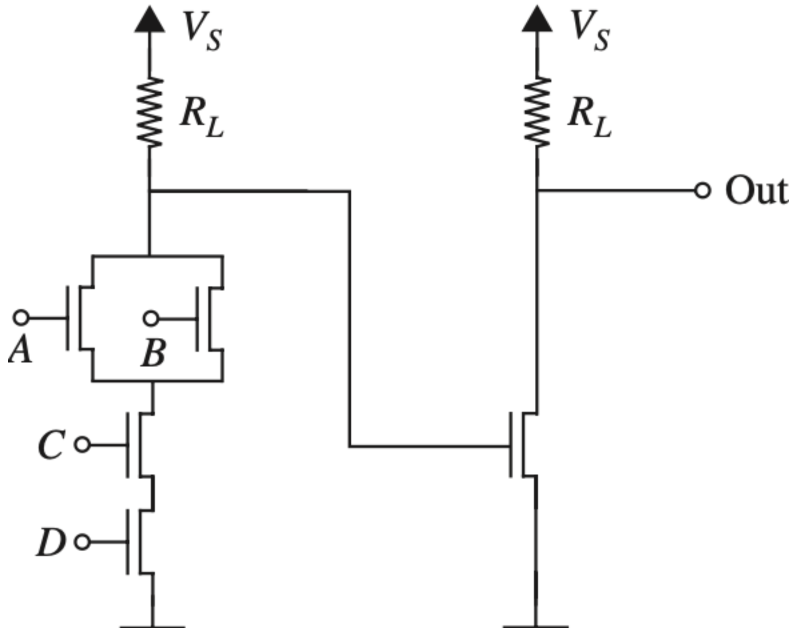
Problem 1

- Analyze the circuits shown below to find the outputs in terms of Boolean inputs A, B, C and D .

i)



ii)

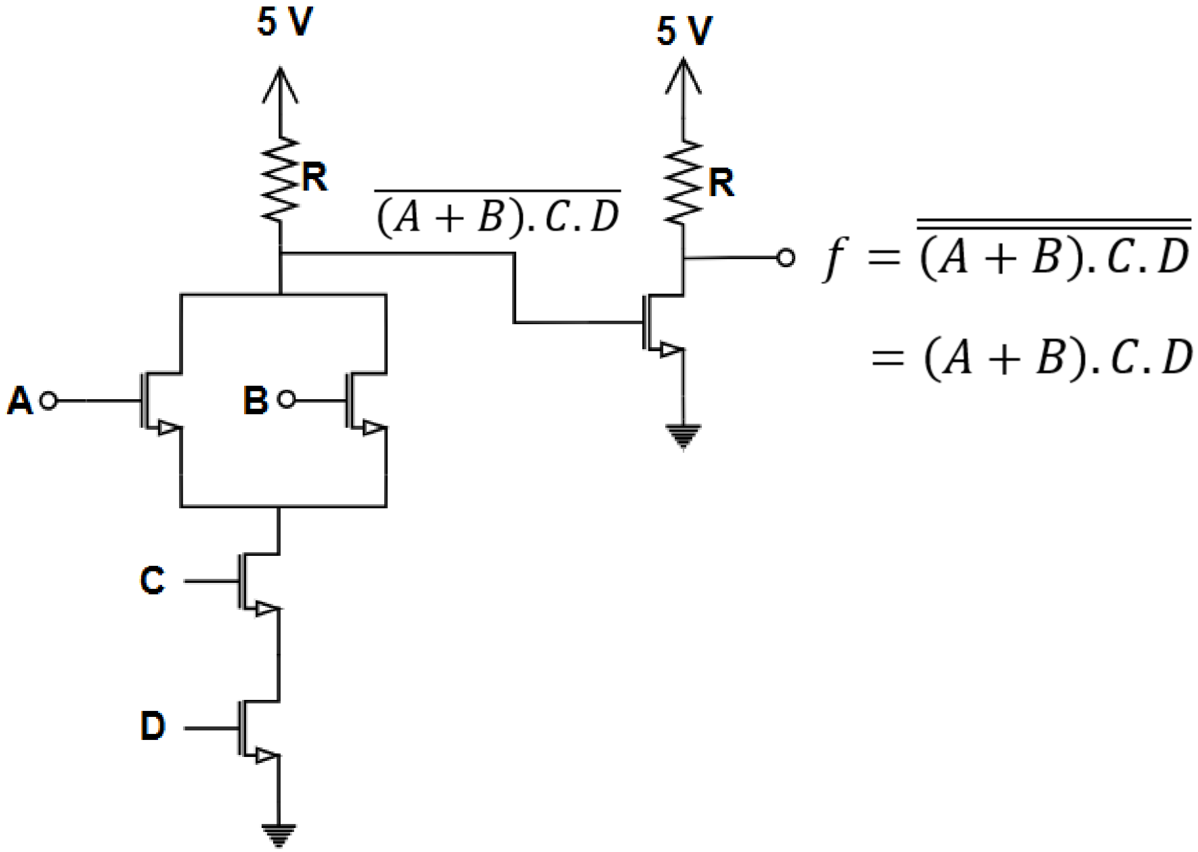


Logic Gate Design with MOSFETs- Examples

Example 1

- Design a circuit using ideal MOSFETs (S-Model) to implement the logic function,

$$f = (A + B).C.D$$

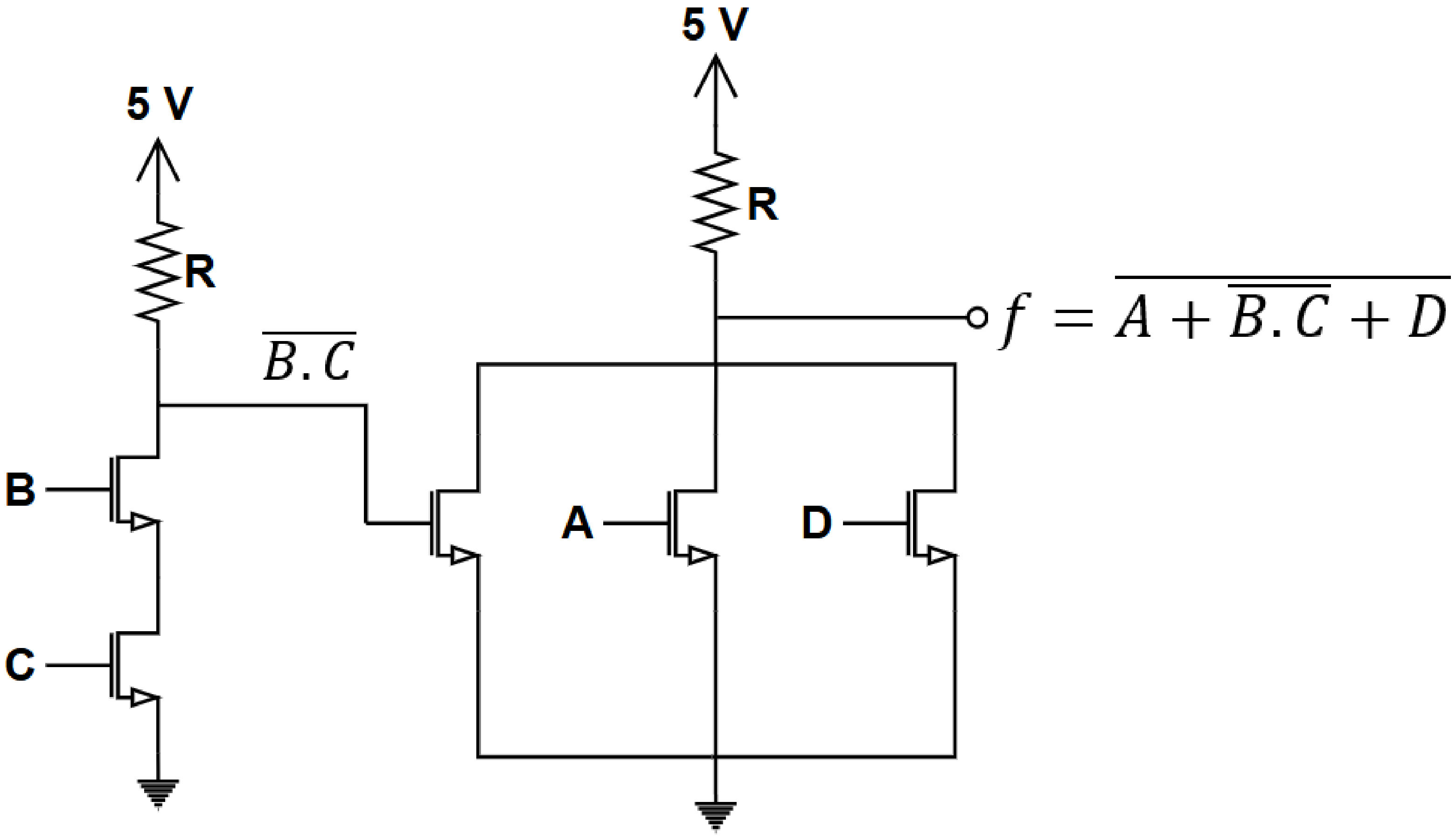


Logic Gate Design with MOSFETs- Examples

Example 1

- Design a circuit using ideal MOSFETs (S-Model) to implement the logic function,

$f = \overline{A + \overline{B.C} + D}$



Practice Problems

- Implement the following logic functions using MOSFETs. Here *A, B, C, D, E* and *F* are Boolean inputs.

i. $f = A.B + C.D$

ii. $f = A + B.C.D$

iii. $f = A + \bar{B}.C$

iv. $f = \overline{(A + B).C + \bar{B}.D}$

v. $f = (\overline{A + B}).C.D$

vi. $f = \overline{A.B} + \overline{C + D}$

vii. $f = A \oplus B$

viii. $f = (AB + C).D.(E + F)$